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AI Applications and Case Histories — ITAI 2372

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A06: AI in Entertainment and Media

Industry Overview

The entertainment and media industry is responsible for creating, distributing, and delivering content such as films, television programs, music, video games, news, and digital media. It serves billions of people worldwide by providing information, storytelling, education, and entertainment. As digital platforms continue to expand, audiences have access to more content than ever before, creating new opportunities and challenges for both creators and media companies.

Artificial intelligence has become one of the most influential technologies shaping this industry. For decades, technology primarily helped distribute content more efficiently. Today, AI is increasingly involved in creating content, recommending what audiences should watch or listen to, and even preserving digital identities long after a person's death. In many ways, AI is no longer working only behind the curtain of the entertainment industry — it is beginning to step onto the stage itself. As a result, AI is changing not only how content is delivered, but also how creativity, storytelling, and audience engagement are experienced.

Current AI Trends

One of the most significant applications of AI in entertainment is AI-assisted content creation. Tools such as AIVA and Amper Music use machine learning algorithms to analyze musical patterns and generate original compositions for films, games, and digital media. Similarly, generative AI models such as GPT-4 can assist with writing, while technologies like Generative Adversarial Networks (GANs) support the creation of digital artwork and design concepts. These tools reduce production time, lower barriers to entry, and allow creators to experiment with new ideas. Rather than replacing human creativity, AI is increasingly acting as a creative partner that helps artists move from inspiration to execution more quickly.

A second major trend is the use of recommendation systems to personalize user experiences. Platforms such as Netflix, Spotify, and Amazon Prime rely on AI algorithms to analyze user behavior and recommend content based on viewing, listening, and purchasing patterns. In a world where thousands of songs, movies, podcasts, and videos compete for attention every day, recommendation systems have become the invisible curators of modern entertainment. Much like a trusted friend

recommending a movie or playlist, these systems help users discover content they might otherwise never find. Features such as Spotify's Discover Weekly and Netflix's personalized thumbnails improve user satisfaction while increasing engagement and retention for companies.

Perhaps the most fascinating and controversial trend is the rise of digital humans and digital resurrection technologies. AI-powered digital clones are now being used to recreate deceased celebrities and preserve historical figures through advanced CGI, voice synthesis, and deepfake-like technologies. One notable example is the planned use of a digital version of James Dean in the film *Back to Eden*. Similarly, AI technology was used to help restore John Lennon's voice for the Beatles' final song, *Now and Then*. Beyond entertainment, companies in China are developing digital replicas of deceased loved ones that can engage in basic conversations with family members. These developments illustrate how AI is expanding beyond content creation into questions of identity, memory, and digital immortality. What once belonged to science fiction is increasingly becoming a real-world discussion about whether a person's voice, image, and legacy can continue to exist long after they are gone.

Ethical Implications

Despite its benefits, AI raises several important ethical concerns within the entertainment and media industry. One significant issue involves privacy and data use. Recommendation systems rely on large amounts of personal data, including viewing habits, listening preferences, search histories, and online behavior. While this information enables highly personalized experiences, it also raises concerns regarding data collection, user consent, and how personal information is stored and utilized by media platforms. The same technologies that help users discover relevant content also create detailed digital profiles of individual behavior.

Another important concern relates to bias, fairness, and transparency. Recommendation algorithms influence which songs become popular, which creators gain visibility, and which content reaches audiences. If these systems favor certain creators, genres, or viewpoints, they may unintentionally limit diversity and reinforce existing biases. Furthermore, most users do not fully understand why specific content is recommended to them. This lack of transparency can create "filter bubbles," where individuals are repeatedly exposed to similar content while alternative perspectives remain hidden. In this sense, recommendation systems do not simply reflect audience preferences — they can also shape them.

Job displacement is also a growing concern as AI becomes increasingly capable of generating music, artwork, written content, and digital performances. Artists, writers, designers, and voice actors have expressed concerns that AI-generated content may reduce opportunities for human creators. In

particular, voice actors worry that digital recreations of iconic voices could continue working indefinitely, potentially limiting opportunities for future generations of performers. While AI may create new technical and creative roles, the industry will need to adapt through workforce reskilling and clear guidelines regarding the use of AI-generated content.

Finally, digital resurrection technologies raise complex questions about consent and ownership. If a deceased individual can no longer approve how their image, voice, or likeness is used, who should make those decisions? Current laws remain unclear in many jurisdictions, creating uncertainty regarding the rights of both celebrities and ordinary citizens. As AI continues to blur the boundary between life, memory, and digital representation, new legal and ethical frameworks will likely become necessary.

Future Trends

Over the next five to ten years, AI is likely to become an even more integrated part of entertainment and media production. Rather than replacing human creators entirely, AI will increasingly function as a collaborative partner capable of generating ideas, assisting with editing, composing music, and creating visual effects. This collaboration may allow creators to focus more on storytelling, artistic direction, and audience engagement while AI handles repetitive production tasks.

Entertainment experiences are also expected to become significantly more personalized. Future recommendation systems may move beyond suggesting existing content and begin generating customized experiences tailored to individual preferences. Audiences may encounter interactive stories, personalized music, and adaptive digital experiences that evolve in real time based on user behavior. The future of entertainment may feel less like choosing from a catalog and more like participating in an experience designed specifically for each user.

The concept of digital immortality may also become increasingly common. Advances in voice cloning, avatar technology, and conversational AI could allow historical figures, celebrities, and even ordinary individuals to maintain digital presences long after their deaths. While these technologies may preserve memories and create new forms of storytelling, they also raise important questions about identity, consent, and the commercialization of human likeness.

As these innovations continue to develop, society will need to balance technological possibilities with ethical responsibility. The future of entertainment will likely depend not only on what AI can create, but also on how responsibly those capabilities are used.

Personal Reflection

What interested me most was not the possibility of AI creating songs or movies, but the fact that it is beginning to influence how people preserve memories and interact with the past. Before studying these materials, I primarily viewed AI as a tool for automation, efficiency, and decision-making. I had never seriously considered the possibility that AI could preserve a person's voice, appearance, or personality long after their death.

The examples involving James Dean, AI-generated replicas, and digital avatars of deceased family members raised questions that felt more philosophical than technical. While many discussions about AI focus on what machines can create, these examples focused on something much more personal: what happens when technology begins to reshape memory itself. The idea that someone could continue to appear in films, conversations, or digital spaces decades after their death is both fascinating and unsettling.

Perhaps the most surprising lesson from this module is that the future of AI in entertainment is not only about creating new content. It is also about changing how people remember, preserve, and interact with the past. For centuries, stories have been one of the primary ways people preserved their legacy. AI introduces a new possibility: preserving not only the story, but the storyteller. Whether that future is inspiring, unsettling, or somewhere in between remains one of the most interesting questions raised by this technology.

References

Course Materials:

- ITAI 2372 Module 06: AI in Entertainment and Media.
- *How AI is Bringing Film Stars Back from the Dead* (BBC Future, 2023).
- Generative AI's Impact on Music.
- *Deepfakes of Your Dead Loved Ones Are a Booming Chinese Business*. MIT Technology Review (2024).

Author's Note:

Google NotebookLM and ChatGPT were used as learning and research support tools throughout the development of this assignment. These tools assisted with organizing course materials, exploring concepts, and deepening understanding of topics discussed in class, including AI advancements, ethics, governance, and future trends. They were used to support the learning process rather than as primary sources of information.